A Practical Manual on

Soil Fertility Management

Based on ICAR 6th Dean Committee Syllabus









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PREFACE

Soil fertility management plays a vital role in sustaining agricultural productivity and maintaining soil health. Understanding the availability and management of essential plant nutrients is fundamental for improving crop yield and ensuring sustainable farming practices. Practical knowledge of soil and plant nutrient analysis helps students develop scientific skills necessary for evaluating soil fertility and making appropriate nutrient management decisions.

This practical manual, **"A Practical Manual on Soil Fertility Management"**, has been prepared in accordance with the guidelines of the Indian Council of Agricultural Research (ICAR) 6th Deans' Committee Report. The manual is designed to provide systematic laboratory exercises for undergraduate students in soil science and agricultural chemistry.

The experiments included in this manual cover important analytical techniques such as the determination of available nitrogen, phosphorus, potassium, sulphur, calcium, magnesium, micronutrients, and organic carbon in soils, as well as nutrient estimation in plant samples. Standard analytical methods like the Kjeldahl Method and other widely accepted procedures have been incorporated to ensure accuracy and practical relevance.

The step-by-step procedures, calculations, and interpretations presented in this manual will help students gain hands-on experience with analytical instruments and laboratory techniques. It is hoped that this manual will serve as a useful guide for students, teachers, and researchers involved in soil fertility and nutrient management studies.

Any constructive suggestions for improvement will be highly appreciated and considered in future editions.

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ABOUT THE PRACTICAL MANUAL

A Practical Manual on Soil Fertility Management (As per ICAR 6th Deans' Committee Report) is designed to provide students with hands-on training in soil and plant nutrient analysis. The manual follows the practical syllabus recommended by the Indian Council of Agricultural Research (ICAR) under the 6th Deans' Committee Report and aims to strengthen students' practical understanding of soil fertility and nutrient management.

The manual includes a series of laboratory exercises covering the principles and use of analytical instruments and standard procedures for determining essential nutrients in soil and plant samples. Experiments include the estimation of available nitrogen, phosphorus, potassium, sulphur, calcium, magnesium, and micronutrients such as zinc, iron, manganese, and copper. It also provides practical methods for analyzing nutrient content in plant materials and organic sources like urea and farmyard manure.

Standard analytical methods such as the Subbiah and Asija Method for available nitrogen, Metson Method for potassium determination, the Kjeldahl Method for nitrogen analysis, and procedures described by M. L. Jackson are incorporated.

Through step-by-step experimental procedures, calculations, and interpretations, this manual helps students develop laboratory skills and understand the role of soil fertility in sustainable crop production. It serves as an essential reference for undergraduate and postgraduate students of soil science, agronomy, and agricultural chemistry.

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DISCLAIMER

The information provided in this manual is intended solely for educational and academic purposes. While every effort has been made to ensure accuracy and reliability, the authors and publishers do not assume any responsibility for errors, omissions, or any consequences arising from the use of the information contained herein.

All experimental procedures should be conducted in properly equipped laboratories under the supervision of qualified personnel. Users must follow standard safety protocols and institutional guidelines while handling chemicals, reagents, and analytical instruments.

The methods described in this book are based on standard scientific procedures; however, slight variations may occur depending on laboratory conditions and equipment. The authors are not liable for any damage, injury, or loss resulting from the application of the techniques described.

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EXPERIMENT - 1

OBJECTIVE: INTRODUCTION OF ANALYTICAL INSTRUMENTS AND THEIR PRINCIPLES

Analytical instruments are devices which are used to measure the physical or chemical properties of assayed substance.

Basic concepts in Instrumental analysis

COLOREMETRY

In coloremetry, color reactions using appropriate reagents are made use of to measure concentration of a substance. Higher the concentration of the substance being determined, greater is the intensity of the color. The intensity of color is measured using an instrument called photoelectric colorimeter.

Coloremeter deals with the determination of concentration of the substance in solution by measurement of the relative absorption transmittance of light with respect to a known concentration of the substance.

About the instrument:

The photoelectric colorimeter is an instrument which helps to determine the extent of absorption of light of particular wavelength through a coloured solution. It consists of

1) A light-source

2) Collimating lens

3) Filter to isolate the required band of wavelength

4) Sample holder/cuvette

5) Photoelectric cell-It convert light energy into electric energy

6) Colorimeter-measures the current output

Operations:

1) See that a proper filter is in position

2) Check whether the galvanometer needle is on central line of galvanometer scale. If not bring it to null position by adjusting the mechanical zero adjustment knob located on top of the instrument (without switching on the instrument)

3) Switch on the instrument (main lamp) and allow it to warm up for 30 minutes (after filling with blank or distilled water) in the holder. Clear the outside with tissue paper and place it in a sample holder.

4) Place the potentiometer needle at zero reading of the scale & see that galvanometer arresting switch (side switch) is in ON position

5) Adjust the reference knob (which adjust the slit across the reference beam) till the deflected needle is brought to null position

6) Switch off the galvanometer side switch remove the cuvette with reagent blank, clean it with distilled water & fill it (appr.5 ml) with standard solution, wipe the outer surface & place it in the instrument

7) The galvanometer side switch is ON & needle of galvanometer deflects from central line to right

8) Rotate the potentiometer knob (located in front of the instrument) anticlockwise until the deflected needle is brought back to central line / null position of the galvanometer. Note down the reading Switch OFF the galvanometer side switch & potentiometer knob is brought back to zero position.

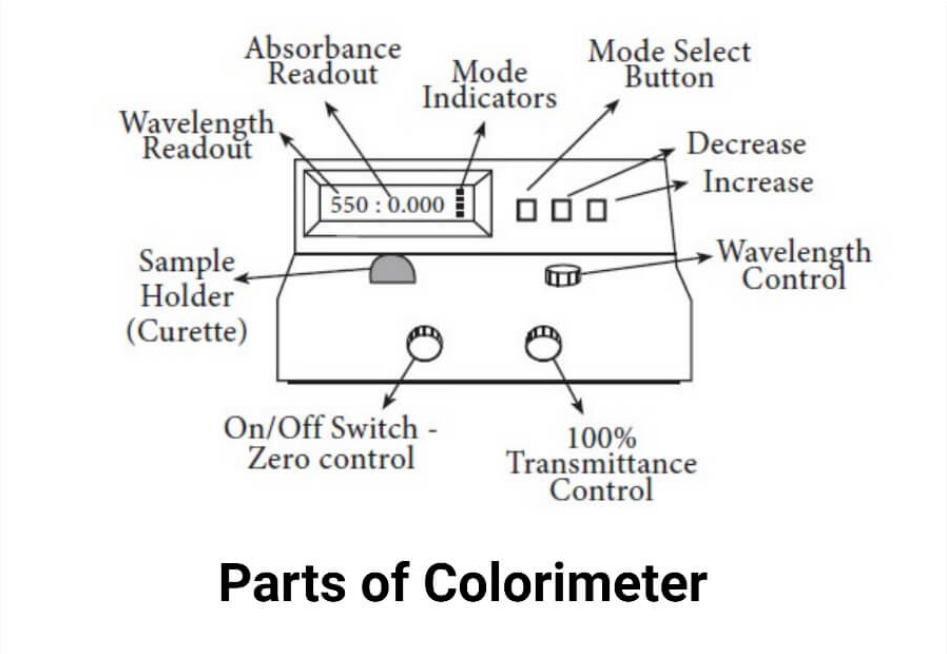
9) Take further readings of all standards using above steps

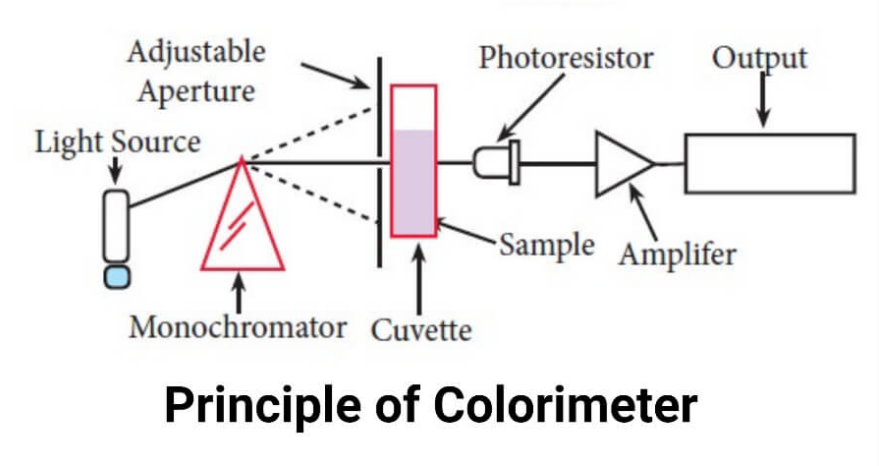
10) When all the readings are over, clean the sample folder & fill it with distilled water. Before it is inserted, switch OFF the main lamp

11) Plot the readings in a graph paper & connect the points by a straight line (standard curve)

12) Find out concentration of sample from the standard curve

Diagram of the Instrument:





FLAME PHOTOMETRY

Flame Photometry makes use of the characteristic radiation given out by different elements when their atoms are excited in a flame. When a fine atomized spray of a solution of a compound containing the element is introduced into the flame atoms of elements absorbs thermal energy from the flame & get excited. As a result extra nuclear electrons are raised to higher energy level, fall hack to original lower energy level energy initially absorbed given off in the form of radiation of discrete wavelength. The amount of radiation given out in this manner will be proportional to the conc. of atoms in the flame. When all other things will be equal the concentration of atoms in the flame will depend upon the conc. of elements in the solution. In flame Photometry the characteristic radiation is isolated using a filter & intensity of isolated radiation is measured by a suitable mechanism such as photoelectric cell &galvanometer.

Essential parts of a Flame Photometer:

1) Pressure regulator for the fuel, gas &air

2) Atomizer:- Aspirates the solution & atomize it in the form of mist

3) Burner:- Produce desired flame in which atoms are excited

4) The optical system:- Collects light energy, renders it monochromatic by a filter or monochromatic

5) Photosensitive detectors:- usually a barrier layer cell converts light energy into electrical energy

6) Galvanometer:- Measure electric output

7) The fuel gas ordinary LPG gas air combination can excite only elements like Li, Na, and K while Ca, Mg requires higher flame temperature (3000°C) for excitation and hence acetylene air combination is required.

Operating Flame photometer:

1) Switch ON the instrument, warm up to 30minutes

2) Insert proper filter

3) Start air compressor & adjust flow rate (0.48-0.6 $kgcm^2$)

4) Place beaker containing distilled water for aspiration

5) Open gas cylinder & adjust gas/flow regulator, light the burner

6) Adjust the regulator to obtain a non-luminous flame with maximum blue cone

7) Aspirate the blank or distilled water

8) Adjust set zero knob of galvanometer to read zero

9) The most conc. standard is aspired first & reading is adjusted to 100 using sensitivity knob

10) Several standards of lower concentration are introduced & reading noted

11) Aspire the blank or distilled water, aspire the unknown solution and note down the reading.

The conc. of unknown solution is calculated from the standard curve prepared with reading on x--axis & conc. on y axis. Critically the curve should be straight line. At the end of determination the following steps are taken to switch OFF flame potentiometer.

1. After aspiring all the samples turn OFF gas supply

2. Switch OFF galvanometer

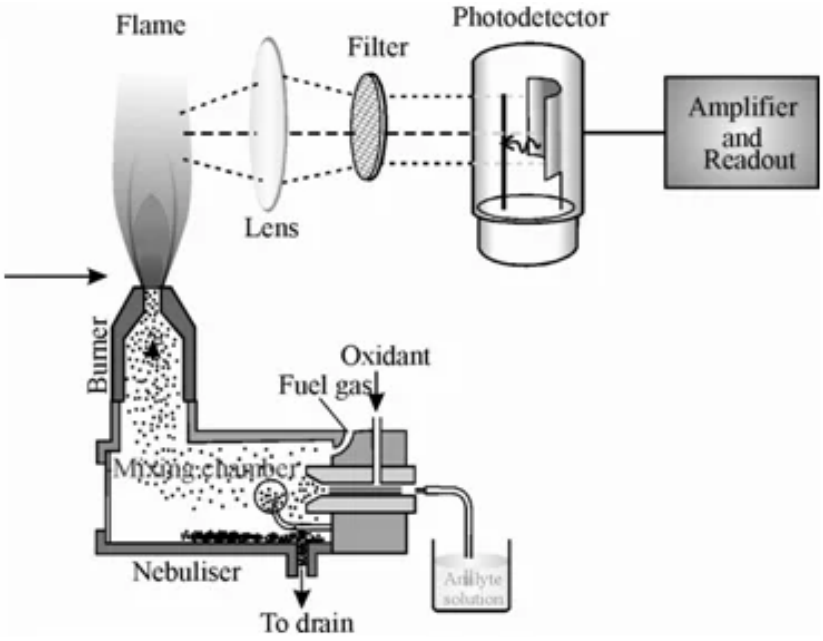
3. Run distilled water & allow to flush out the atomizer & burner for sometimes

4. Remove the distilled water & allow only compressed air to flow

5. Switch OFF the compressor after 5 min to cool the instrument

Diagram of the instrument:





EXPERIMENT- 2

OBJECTIVE: DETERMINATION OF AVAILABLE NITROGEN IN SOILS (SUBBIAH AND ASIJA, 1956)

Principle

A known weight of the soil is mixed with excess of alkaline KMnO4 and distilled. Ammoniagas released by distillation is absorbed in a known volume of standard sulphuric acid excess ofwhich is titrated with standard alkali using methyl red as an indicator.

1. Decomposition of potassium permanganate

2. Oxidation of organic matter (general form)

$$\text{Organic matter} + O \rightarrow R - NH_2$$

3. Hydrolysis / decomposition of amine

$$R - NH_2 + H_2O \rightarrow R - OH + NH_3$$

4. Reaction of ammonia with sulfuric acid

$$2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$$

5. Neutralization reaction

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$**Apparatus**

Distillation apparatus, distillation flask, glass beads, 250 ml beakers, conical flask, burette and pipette etc.

Reagent

- **0.32% $KMnO_4$:** Dissolve 3.2 g of $KMnO_4$ in distilled water and make the volume to one litre.
- **2% Boric acid:** Dissolve 20 g of boric acid powder in warm water by stirring and dilute to one litre.
- **2.5% NaOH:** Dissolve 25 g of sodium hydroxide pellets in water and make the volume to one litre.

- **0.1 M Potassium Hydrogen Phthalate:** Dissolve 20.422 g of the salt in distilled water and dilute to one litre. This is a primary standard and does not require standardization.
- **0.02 M H_2SO_4:** Prepare approximately 0.1 M H_2SO_4 by adding 5.6 mL of concentrated H_2SO_4 to about one litre of distilled water. From this, prepare 0.02 M H_2SO_4 by diluting a suitable volume (20 mL made to 100 mL) with distilled water. Standardize it against 0.1 M NaOH solution.
- **0.1 M NaOH:** Dissolve 4 g NaOH in 100 mL distilled water. Standardize against potassium hydrogen phthalate.
- **Mixed Indicator:** Dissolve 0.066 g of methyl red and 0.099 g of bromocresol green in 100 mL of ethyl alcohol. Add 20 mL of this mixed indicator to each litre of 2% boric acid solution.

Procedure

- Measure **20 mL of 2% boric acid containing mixed indicator** in a **250 mL conical flask** and place it under the receiver tube. Carefully dip the receiver tube in the boric acid solution.
- Take **2.5 g of soil sample** in a **350 mL distillation tube***(run a blank without soil in a 350 mL distillation tube before starting sample)*.
- Moisten the soil with about **10 mL of distilled water**, and wash down the soil, if any, adhering to the neck of the flask.
- Add **25 mL (Soil : 0.32% $KMnO_4$ :: 1 : 5)** of **0.32% $KMnO_4$ solution** in the distillation tube manually or instrumentally.
- Add a few **glass beads or broken pieces of glass rod**.
- Add **2–3 mL of paraffin liquid**, avoiding contact with the upper part of the neck of the flask.
- Add **25 mL of 2.5% NaOH solution** and immediately attach to the rubber stopper fitted in the alkali trap.
- Press the **run switch** and continue distillation until about **100 mL of distillate** is collected.
- First remove the conical flask containing distillate and then remove the distillation tubeto avoid **back suction**.
- Titrate the distillate against **0.02 M H_2SO_4** taken in burette until **pink colour starts appearing**.
- Carefully remove the distillation tube and drain the contents in the sink.

Note

If brown colour is not appearing in the distillation tube during distillation (after 6–7 minutes), then add ***5–10 mL of 2.5% NaOH solution***.

Observation and calculation

1. Weight of soil = 20 g

2. Volume of N/50 H_2SO_4 taken (X) = 20 ml

3. Volume of N/50 NaOH used (Y) = y ml

4. Volume of N/50 H_2SO_4 used for NH, (X-Y) = ml

5. 1 ml of N/50 H_2SO_4 = 0.00028 g N

6. Per cent available nitrogen (X-Y) x 0.00028 × 100/20

7. Available nitrogen in ppm = percentage x 10

8. Available nitrogen Kg/ha (X-Y) x 0.00028/20 × 2240000 or = (X-Y) x 28 x 1.12 = (X-V) x 31.36

Precautions

1. Dip the delivery tube end in the receiver containing standard acid before adding NaOH solution in the distillation flask

2. Connect the distillation flask to the distillation apparatus immediately after the addition of NaOH to avoid the loss of ammonia.

3. During distillation, first remove the receiver flask and then heater to avoid back suction of the distillate.

4. Collect about 100 ml of distillate in 30 minutes steady distillation.

EXPERIMENT- 3

OBJECTIVE: DETERMINATION OF AVAILABLE PHOSPHORUS IN SOIL

Principle

Soil phosphorus is extracted using an appropriate extract ant depending on the soil pH. The extracted solution is then treated with an acid-molybdate reagent, leading to the formation of a phospho-molybdate complex. This complex is subsequently reduced using reducing agents such as hydrogen sulphide (H_2S), ascorbic acid, or stannous chloride ($SnCl_2$), resulting in the development of a blue-colored compound known as molybdenum blue. The intensity of this blue color is directly proportional to the phosphorus concentration present in the sample.

The reaction involved in the formation of the heteropoly complex is:

$H_3PO_4 + 12\ H_2MoO_4 \rightarrow H_3P(Mo_3O_{10})_4 + 12\ H_2O$

In this heteropoly compound, molybdenum exists initially in the +6 oxidation state (Mo^{6+}), which is reduced to lower oxidation states (Mo^{3+} or Mo^{5+}) by the reducing agents. The presence of unpaired electrons in the reduced molybdenum species leads to the characteristic blue color, which shows maximum absorbance at 660 nm.

In soils, phosphorus is primarily present as orthophosphate in various chemical forms, but only a small fraction is readily available for plant uptake. Therefore, assessing available phosphorus is crucial for evaluating soil fertility. Several extract ants have been developed to estimate available phosphorus, including water, neutral salt solutions (KCl, $CaCl_2$), strong mineral acids (e.g., HCl, HNO_3, H_2SO_4), weak organic acids (citric, lactic, acetic acids), buffered solutions (such as Truog's, Egner's, and Morgan's solutions), and alkaline extract ants (NaOH, K_2CO_3, $NaHCO_3$). Additionally, chelating agents like EDTA and DTPA may be used to extract specific forms of bound phosphorus.

Among the various methods, two widely used procedures are Bray's Method No. 1 for acidic soils and Olsen's Method for neutral to alkaline soils.

1.Bray's method No. 1 (Bray and Kurtz, 1945) for acid soils

The estimation of specific categories of soil P considered to be of significance to crop growth has lead to the introduction of neutral NH_4F (ammonium fluoride) solution as an extracting for adsorbed phosphate and NH_4F-HCl combination for adsorbed plus acid soluble form. This method is primarily meant for soils which are moderately to strong acid. The Bray's No. I reagent consists 0.03N NHF in 0.025 N HC1.

Apparatus and Instruments:

- Spectrophotometer
- Shaker
- Pipettes
- Beakers
- Conical flasks
- Volumetric flasks etc.

Reagents:

Bray Extractant No 1 (0.03M NH_4F in 0.025M HCI): Dissolve 1.11 g of NH_4F (AR) in one litre of 0.025N HCI.

Dikeman Bray's reagent (Molybdate reagent):

Dissolve 15.0 g $(NH_4)_2MoO_4$ in 300 mL warm distilled water, cool and add the solution to 350 mL.of 10N HCl solution gradually with stirring. Dilute to one litre with distilled water.

Stannous chloride solution (40% Stock Solution):

Dissolve 10 g $SnCl_2.2H_2O$ in 25 ml. of concentrated HCl. Add a piece of pure metallic tin and store the solution in a glass stoppered bottle.

Stannous chloride solution (Working Solution):

Dilute 1 ml. of the stock solution of stannous chloride to 66.0 mL with distilled after just before use. Prepare fresh dilute solution every working day.

Preparation of the Standard Curve:

Dissolve 0.439 g of pure dry KH_2PO_4 in about half a litre of distilled water. About 25 ml. 7N H_2SO_4 is added and made up to one litre with distilled water. This solution contains 100 ppm stock solution of P (100 mg P/L). Preserve this as a stock standard solution of phosphate. From this, a 2 ppm Psolution is made. Take 0,1, 2, 3, 4, 5 and 6 mL of this 2 ppm solution to 25 mL. volumetric flasks. To these 5 mL of extracting reagent (Bray's) is added as described above by adding Bray's No. 1, 5 ml. of Dickman Bray's reagent, 1 mL $SnCl_2$ and make the volume with distilled water and take reading as per sample readings.

Procedure:

- Take 2.5 g of soil and 25 mL of the Bray's reagent (1:10 Soil: Solution) are shaken for 5 minutes in 250 mL conical flask and filtered.

- Take 5 mL of the filtered soil extract with a bulb pipette in a 25 ml. volumetric flask. To avoid interference of fluoride 7.5 mL. of 0.8 M boric acid (50g H_3BO_3) per litre) can be added to 5 mL of the extract.
- Add 5 mL of the molybdate reagent and add about 10 mL distilled water, shake and add 1 ml. of the dilute $SnCl_2$ solution with a pipette. Make up the volume with distilled water and shake thoroughly.
- Read the blue colour after 10 minutes on the spectrophotometer at 660 nm wavelength after setting the instrument to zero with the blank prepared similarly.

Phosphorus is determined spectrophotometrically by Dickman and Bray's (Dickman and Bray, 1940) method.

Olsen's method (Olsen et al., 1954) for neutral and alkaline soils

A 0.5M $NaHCO_3$ solution adjusted to pH 8.5 has been designed to control the ionic activity of Ca through the solubility product of $CaCO_3$ in case of neutral and calcareous soils. By this process, the most reactive form of P is extracted from the phosphates of iron, aluminium and calcium present in the soil.

Apparatus:

- Spectrophotometer
- Shaker
- Pipettes
- Beakers
- Conical flasks
- Volumetric flasks etc

Reagents:

- **0.5 M $NaHCO_3$:** Dissolve 42 g Sodium bicarbonate in 1 litre of distilled water and adjust the pH to 8.5 by addition of dilute NaOH or HCl. Filter it, if necessary.

Activated carbon-Darco G 60 or P free charcoal.

- **Moly date reagent:** Same as for the Bray's Method No. I except that use 400mL of 10N HCI instead of 350 mL/litre.
- **Stannous chloride solution:** Same as in Bray's Method No. 1.
- **Preparation of the Standard Curve:** Dissolve 0.439 g of pure dry KH_2PO_4 in about half a litre of distilled water. About 25 ml. 7N H_2SO_4 is added and made up to 1 litre with distilled water. This solution contains 100 ppm stock solution of P (100 mg P/L). Preserve this as a stock standard solution of phosphate. From this, a 2 ppm P solution is made. Take 0,1, 2, 3, 4, 5 and 6 mL of this 2 ppm solution to 25 mL. Volumetric flasks. To these 5 mL. of extracting reagent (Olsen's) is added as described above by

adding, 5 ml. of molybdate reagent (for Olsen's method), 1 mL $SnCl_2$ and make the volume with distilled water and take the colour reading on the spectrophotometer at 660 nm wavelength. Plot the absorbance reading against P mg/L and prepare the standard curve by adjoining points.

Procedure:

- Weigh 2.5 g soil sample in a 250 mL. conical flask.
- Add 1-2 g of Darco G 60 or P free charcoal and 50 mL of the bicarbonate extract ant (0.5M $NaHCO_3$, pH8.5).
- Shake for 30 minutes on the mechanical shaker and filter through Whatman No. 42 filter paper
- Transfer 5 mL of filtrate in to 25 ml. volumetric flask and gradually add 5 ml. of ammonium molybdate containing 400 ml. of 10NHCL
- Stir slowly and carefully to drive out the CO_2evolved.
- After stop of bubbliness, add distilled water washing down the sides and bring the volume to 22ml.

Add 1 mL of freshly prepared diluted $SnCl_2$, shake a little and make up the volume.

- ❖ Run a blank without soil in similar manner.
- ❖ Read the blue color after 10 minutes on the spectrophotometer at 660 nm wavelength (or red filter for colorimeter) after setting the instrument to zero with the blank prepared similarly.

Precautions:

1. Clean all glassware with distilled water.

2 Start filtration quickly after shaking

3. Before taking reading, warm up the instrument minimum 30min.

4 Take reading after 10min.

Observation:

SI	Spectrophotometer reading (Abs)

Calculation:

Available P (kg/ha) =R dilution factor × 2.24

Where, R ppm P in the sample (read from standard curve)

Dilution factor = 50/2.5×25/5 = 100

Weight of the soil taken = 2.5 g.

Volume of the extract = 50 mL;

Volume of the aliquot taken for estimation = 5 mL

Volume made for estimation (dilution 5 times) = 25 mL

Ratings:

P kg/ha	**Low**	**Medium**	**High**
Bray's Method	**< 30**	**30-60**	**>60**
Olsen's Method	**< 15**	**15-30**	**>30**

EXPERIMENT- 4

OBJECTIVE: DETERMINATION OF AVAILABLE POTASSIUM IN SOIL (METSON, 1956)

Principle

The term **available potassium** represents both the **exchangeable** and **water-soluble** forms of potassium present in the soil. The readily exchangeable and water-soluble forms of potassium are determined by extracting the soil with **neutral normal ammonium acetate (NH_4OAc)** solution.

The **ammonium ion (NH_4^+)** replaces potassium (K^+) from the soil exchange complex rapidly and effectively. Other cations may also replace potassium, but their action is slower and may vary depending on the duration of contact.

Thus, ammonium acetate provides a **sharp and efficient extraction** of available potassium from the soil.

The potassium present in the extract is then estimated using a **flame photometer**, which measures the intensity of light emitted by potassium ions.

$$(\text{Colloid}) - K^+ + NH_4CH_3COO \rightarrow (\text{Colloid}) - NH_4^+ + KCH_3COO$$

The estimation of potassium in the extract is carried out with the help of flame photometer.

Apparatus

Flame photometer, shaking machine, conical flasks, volumetric flasks, pipette, beaker, funnel etc.

Reagents

1. 1N neutral ammonium acetate solution: Take 700 ml of distilled water in a one litre volumetric flask. Add 57 ml of glacial acetic acid (99.5%) and then add 69 ml of concentrated ammonium hydroxide. Dilute it to about 900 ml and adjust the pH at 7.0 by adding 3 N NH_2OH or 3 N CH_3COOH and make up to one litre. It can also be prepared by dissolving 77.04 g of ammonium acetate in 900 ml of distilled water and adjust the pH at 7.0 by adding NH_4OH or CH_3 COOH as required and make the volume one litre.

2. Standard K solution: Dissolve 1. 908 g AR grade KCI in distilled water anti make the volume one litre. This solution contains 1 mg K/ml Le. 1000 ppm K.

3. Working solution of K: Take 10 ml of 1000 ppm K solution in 100 m1 volumetric flask and make the volume. This will contain 100 ppm K.

Procedure

1. Take 5 g soil in 150 ml conical flask.

2. Add 25 ml ammonium acetate solution and shake for 5 minutes on a shaker

3. Filter the contents through a Whatman No. 1 filter paper and reject first few drops of the filtrate.

4. Feed the filtrate to flame photometer after necessary setting and calibration of the instrument.

5. Also run a blank without soil using the same procedure,

Preparation of standard curve

From the 100 ppm solution take 10, 15, 20, 25, 30 and 40 ml of K in each 100 ml volumetric flask and make the volume. This will give 10, 15, 20, 25, 30 and 40 ml of K. The concentration of K may be reduced according to the working range of flame photometer. The curve is obtained by plotting the readings of flame photometer against the various concentration of K by setting the reading at zero for the blank and 100 for the standard K solution having maximum concentration to be used.

Observation and Calculation

Weight of soil taken = 5 g 1

Volume of extract = 25 ml 2

3. Reading of flame photometer = X

4. ppm of available K from the curve = R

P pm of available K kg/ha = Rx Volume of extractant (25ml)/ Weightof soil(5g) x2.24

= R x 11.2

Available K_2O Kg/ha = R x 11.2×1.23

Precautions

1. pH of the ammonium acetate solution should be adjusted at 7.0

2. In flame photometer gas and air pressure should not fluctuate in gel steady reading.

3. Few ml of filtrate in the beginning should be rejected.

4. Use the proper filter of K and do not take the reading without filter.

EXPERIMENT 5

OBJECTIVE: DETERMINATION OF AVAILABLE SULPHUR IN SOILS

Principle

Soil is shaken with 0.15% $CaCl_2$ solution. Chloride ions displace adsorbed sulphate during extraction. Calcium ions suppress the extraction of soil organic matter and hence eliminate the contamination caused by extractable organic S. The filtrate is analysed for S by the turbid metric method of Chesnin and Yien (1950) in which the turbidity produced due to precipitation of sulphate as barium sulphate is measured on a Colorimeter at a wavelength of 420 nm using a blue filter. Gum acacia solution is added to stabilize the turbidity so that the precipitate of barium sulphate does not settle down.

Reactions

1. Dissociation of Calcium Chloride

$$CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$$

2. Exchange on Micelle (Colloid)

$$[\text{Micelle}]\, SO_4^{2-} + 2Cl^- \rightleftharpoons [\text{Micelle}]\, Cl^- + SO_4^{2-}$$

3. Dissociation of Barium Chloride

$$BaCl_2 \rightarrow Ba^{2+} + 2Cl^-$$

4. Precipitation Reaction

$$SO_4^{2-} + Ba^{2+} \rightarrow BaSO_4 \downarrow$$

Reagents

1. **Extracting solution:** Dissolve 1. 986 g of calcium chloride dihydrate ($CaCl_2$ $2H_2O$) in distilled water and dilute to one litre.

2 **Barium chloride ($BaCl_2$):** 30 to 60-mesh crystals: Grind barium chloride crystals until they pass through 30-mesh sieve and are retained on a 60-mesh sieve.

3. **Gum acacia solution (0.25%):** Dissolve 0.25 g gum acacia in distilled water and dilute to 100 ml.

4. **Standard sulphur (100 ppm S):** Dissolve 0.5434 g of the reagent grade potassium sulphate (K_2SO_4) in distilled water and dilute to one litre.

Apparatus

- 150-ml Erlenmeyer flask
- Funnel
- Pipettes 25-ml volumetric
- Flasks electric shaker
- Colorimeter and
- Whatman No.42 filler paper.

Procedure

(a) Preparation of standard curve

Pipette out 0.25, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0 and 5.0 ml of 100 ppm S solution in different 25-ml volumetric flasks. Thus sires of standard S solutions of 1, 2, 4, 6, 8, 10, 12 and 20 ppm will be obtained, respectively after making final volume to 25 ml. In each flask add 1.0 g of 30-60 mesh $BaCl_2$ crystals. Swirl for one minute to dissolve the crystals and add 1 ml of 0.25% solution of gum acacia. Make up the volume in each flask with distilled water and shake well. Within 5 to 30 minutes after the development of turbidity, read the standards on a colorimeter at 420 nm using a blue filter. Plot a standard curve showing relationship between concentration of S and turbidity/absorbance readings.

(b) Analysis of test sample

1. Weight 10 g air-dried soil and transfer it to a 150 ml conical flask

2. 60 ml of 0.15% $CaCl_2$ solution and shake for 30 minutes on an electric shaker

3. Filter the suspension through Whatman No. 42 filter paper.

4. Pipette out 20 ml of the filtrate in 25 ml volumetric flask.

5. Add one gram of 30-60 mesh BaCl, and Shake for 1 minute.

6. Thereafter add 1 ml of 0.25% gum acacia solution.

7. Finally make up the volume by adding distilled water and Shake for 1 minute.

8. Within 5 to 30 minutes after development of turbidity take the reading on Spectro calorimeter.

Calculation

Weight of soil taken = 10 g

Volume of extract ant added= 50 ml

First dilution = 5 times

Volume of aliquot taken = 20 ml

Final volume = 25 ml

Second dilution = 1.25 times

Absorbance reading = A

pom of S from standard curve against A value = Y

ppm of S in soil = Yx6.25 ppm

kg of S/ha= ppm x 2.24

Inference: Soil having available S less than 10 ppm are rated as deficient in sulphur.

Precautions:

1. Before taking reading, warm up the instrument minimum 30min.
2. Take reading within 30min

EXPERIMENT - 6

OBJECTIVE: ESTIMATION OF AVAILABLE CA AND MG IN SOILS

Principle

The method described here was developed by Lavkulich (1981) for the standard analysis of a wide range of soil types. It involves fewer procedural steps compared to other similar methods, such as that proposed by McKeague (1978).

Several limitations associated with this method for measuring exchangeable cations and cation exchange capacity (CEC) have been widely discussed in the literature (Chapman, 1965; Bache, 1976; Rhoades, 1982; Thomas, 1982). However, it is generally accepted, as concluded by Thomas (1982), that *there is currently no evidence suggesting that cations other than NH_4^+ provide results that are less arbitrary than those obtained using NH_4^+.*

Errors due to the dissolution of $CaCO_3$ and gypsum will result in an excess of Ca^{3+} being extracted by NH_4^+ and a decrease in the amount of NH_4^+ retained due to competition between Ca^{2+} and NH_4^+ during equilibration in the saturating step. In soils containing these minerals, exchangeable Ca will be too high and total CEC too low. The former problem cannot easily be corrected (Thomas 1982); however, more accurate measurement of CEC in this type of soil can be obtained by using the method described by Rhoades(1982).

Fixation of K and NH_4 in phyllosilicates can result in either an over- or underestimation of exchangeable K when NH_4 is used as an extractant depending on whether the NH4 moves through the interlayer positions replacing the K' or whether it causes the collapse of the edges preventing further exchange.

Compared to the other methods presented in this chapter, this method uses a larger sample size, which helps to decrease the sample to sample variability. Another advantage of this procedure is that there are no decantation steps that can cause the loss of sample, particularly in the case of organic soils.

The method described below can be used to measure either exchangeable cations and CEC or just exchangeable cations. In the latter case, the sum of exchangeable cations (including Al) could be

used as an estimate of CEC. Due to the high pH of the extracting solution, the amount of Al measured will usually be lower than that displaced by $BaCl_2$ or KCI.

Calcium by Versenate (EDTA) method

Apparatus:

- Shaker
- Porcelain dish
- Beakers
- Volumetric/conical flask.

Reagents:

Ammonium chloride:

Ammonium hydroxide buffer solution: Dissolve 67.5 g ammonium chloride in 570 ml of conc. ammonium hydroxide and make to 1 litre.

Standard 0.01N calcium solution:

Take accurately 0.5 g of pure calcium carbonate and dissolve it in 10 ml of 3N HCL. Boil to expel CO_2 and then make the volume to 1 litre with distilled water.

EDTA solution (0.01N):

Take 2.0 g of versenate, dissolve in distilled water and make the volume to 1 litre. Titrate it with 0.01N calcium solution and make necessary dilution so that its normality is exactly equal to 0.01N.

Muroxide indicator powder:

Take 0.2 g of muroxide (also known as ammonium purpurate) and mix it with 40 g of powdered potassium sulphate. This indicator should not be stored in the form of solution, otherwise it getsoxidized.

Sodium diethyl dithiocarbamate crystals: It is used to remove the interference of other metalions.

Sodium hydroxide 4N:

Prepare 16% soda solution by dissolving 160 g of pure sodium hydroxide in water and make the volume to 1 litre. This will give pH 12.

Procedure:

- Take 5 g air dried soil sample in 150 ml conical flask and add 25 ml of neutral normal ammonium acetate. Shake on mechanical shaker for 5 minutes and filter through Whatman No. 1 filterpaper.
- Take a suitable aliquot (5 or 10 ml) and add 2-3 crystals of carbarnate and 5 ml of 16% NaOH solution.
- Add 40-50 mg of the indicator powder. Titrate it with 0.01N EDTA solution till the colour gradually changes from orange red to reddish violet (purple). It is advised to

add a drop of EDTA solution at an interval of 5 to 10 seconds, as the change of colour is not instantaneous.

- The end point must be compared with a blank reading. If the solution is over titrated, it should be back titrated with standard calcium solution and exact volume used is thus found.
- Note the volume of EDTA used for titration.

Observation:

S.N.	Initial burette reading (mL)	Final burette reading (mL)	Final volume (mL)

Calculation

If $N_1$is the normality of $Ca^{2+}$and $V_1$is the volume of aliquot taken, and N_2, $V_2$are the normality and volume of EDTA used respectively, then:

$$N_1V_1 = N_2V_2$$

Or,

$$N_1 = \frac{N_2V_2}{V_1} = \frac{\text{Normality of EDTA} \times \text{Volume of EDTA}}{\text{mL of aliquot taken}}$$

Here, N_1(Normality) = equivalent of Ca^{2+} present in one litre of aliquot.

Hence,

$$Ca^{2+}\ \text{me / litre} = \frac{\text{Normality of EDTA} \times \text{Volume of EDTA} \times 1000}{\text{mL of aliquot taken}}$$

When expressed on soil weight basis:

$$Ca^{2+}\ \text{me / 100 g soil} = \frac{100}{\text{wt. of soil}} \times \frac{\text{extract volume}}{1000} \times \text{Ca as me / litre}$$

Calcium plus Magnesium by Versenate (EDTA) method

Magnesium in solution can be titrated with 0.01N EDTA using Eriochrome black T dye as indicator at pH 10 in the presence of ammonium chloride and ammonium hydroxide buffer. At the end point, colour changes from wine red to blue or green. When calcium is also present in the solution this titration will estimate both calcium and magnesium. Beyond pH 10 magnesium is not bound strongly to Erichrome black-T indicator to give a distinct endpoint.

Apparatus:

- Shaker
- Porcelain dish
- Beakers
- Volumetric/conical flask.

Reagents:

EDTA or Versenate solution (0.01N): Same as in calcium determination. Ammonium chloride-ammonium hydroxide buffer solution: Same as in calcium determination.

Eriochrome black T indicator: Take 100 ml of ethanol and dissolve 4.5 g of hydroxyl amine hydrochloride in it. Add 0.5 g of the indicator and prepare solution. Hydroxylamine hydrochloride removes the interference of manganese by keeping it in lower valency state (Mn^{2+}). Or mix thoroughly 0.5 g of the indicator with 50 g of ammonium chloride.

Sodium cyanide solution (2%) or sodium diethyl dithiocarbamate crystals: This is used to remove the interference of copper, cobalt and nickel.

Procedure:

- Take 5 g air dried soil in 150 ml flask, add 25 ml of neutral normal ammonium acetate solution and shake on a mechanical shaker for 5 minutes and filter through Whatman No.1 filter paper.
- Pipette out 5 ml of aliquot containing not more than 0.1 meq of Ca plus Mg. If the solution has a higher concentration, it should be diluted.
- Add 2 to 5 crystals of carbamate and 5 ml of ammonium chloride-ammonium hydroxide buffer solution. Add 3-4 drops of Eriochrome black-T indicator.
- Titrate this solution with 0.01N versenate till the colour changes to bright blue or green and no tinge of wine red colour remains.

Calculation

If N_1 and V_1 are the normality (concentration of $Ca^{2+} + Mg^{2+}$) and volume of aliquot taken, and N_2, V_2 are the normality and volume of EDTA used respectively, then:

$$N_1 V_1 = N_2 V_2$$

Or,

$$N_1 = \frac{N_2 V_2}{V_1} = \frac{\text{Normality of EDTA} \times \text{Volume of EDTA}}{\text{mL of aliquot taken}}$$

Here, N_1(Normality) = equivalents of $Ca^{2+} + Mg^{2+}$ present in one litre of aliquot.

Hence,

$$(Ca^{2+} + Mg^{2+}) \text{ me / litre} = \frac{\text{Normality of EDTA} \times \text{Volume of EDTA} \times 1000}{\text{mL of aliquot taken}}$$

Magnesium Calculation

$$\text{Milliequivalent (me) of } Mg^{2+} = \text{me}(Ca^{2+} + Mg^{2+}) - \text{me of } Ca^{2+}$$

When expressed on soil weight basis:

$$(Ca^{2+} + Mg^{2+}) \text{ me / 100 g soil} = \frac{100}{\text{wt. of soil}} \times \frac{\text{extract volume}}{1000} \times (Ca^{2+} + Mg^{2+}) \text{ me / litre}$$

Observation:

S.N.	Initial burette reading (mL)	Final burette reading (mL)	Final volume (mL)

EXPERIMENT -7

OBJECTIVE: ESTIMATION OF AVAILABLE ZN, FE, MN, CU IN SOILS

Principle

Diethylene triamine pentaacetic acid (**DTPA**) is a chelating agent used for the determination of available **Fe, Mn, Cu, and Zn** in soil. When the soil is shaken with a DTPA solution, it forms stable, soluble complexes with these metal ions, thereby extracting them into solution.

The presence of **$CaCl_2$** and **triethanolamine (TEA)** in the extracting solution helps to slightly raise and buffer the soil pH. This also minimizes the interference caused by **Ca^{2+}** and **Mg^{2+}** ions.

The micronutrients (**Fe, Mn, Cu, and Zn**) present in the DTPA extract are then determined using an **Atomic Absorption Spectrophotometer (AAS)** with their respective hollow cathode lamps.

Apparatus:

- Atomic Absorption Spectrophotometer
- shaking machine
- Centrifuge
- beaker
- Pipette
- volumetric flask
- conical flask
- funnel etc.

Reagents:

1. DTPA 0.005 M solution

2. TEA 0.1 M (AR or extra pure) solution

3. $CaCl_2.2H_2O$ (AR) 0.01M

4. Dilute HCl (1:1) AR diluted with double distilled water.

The extracting reagent is prepared by taking 1.967 g of DTPA, 1.479 g. $Cacl_2.2H_2O$ and 13.3 mL of TEA in 100 mL of glass or double distilled water and dilute approximately to 900 mL.. Adjust the pH of the solution to 7.3 by adding dilate HCl (1:1), while stirring make the volume of the extracting solution to one litre. The solution remains stable for several months.

Preparation of 100 ppm standard solutions:

Element	Atomic weight	Micro nutrient salt	Molecular weight	Quantity of salt in g for 1 litre solution
Zn	54.38	$ZnSO_4.7H_2O$	287.56	0.4398
Cu	63.54	$CuSO_4.5H_2O$	249.69	0.39259
Mn	57.94	$MnCl_2.4H_2O$	197.69	0.3602
Fe	55.85	$FeSO_4.7H_2O$	278.02	0.4977

From the above 100 ppm solution a 10 ppm working standard solution is prepared for Mn", Cu and Zn by taking 10 mL of 100 ppm solution in a 100 mL volumetric flask and making the volume 100 mL with the help of glass distilled water. From the 10 ppm solution take 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10 mL in 50 mL volumetric flask and make up the volume.

This will give 0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8 an d 2.0 ppm of respective micronutrient solution. For iron take direct 0, 1, 2, 3, 4, 5, 10 and 12 mL of 100 ppm solution in 50 mL volumetric flask and make up the volume with glass distilled water, this will give 0, 2, 4, 6, 8, 10, 16, 20 and 24 ppm iron solution.

Procedure:

1. Take 10 g of soil in a 250 mL conical flask and add 20 mL. DTPA extracting solution toit and stopper it well.

2. Shake the conical flask for 120 minutes, using mechanical shaker or centrifuge it using centrifuge tube and filter the contents through Whatman No. 42 filter paper.

3. Prepare a blank by adding all the solution except soil by shaking and filtration.

4. Take reading directly by AAS, using acetylene gas and hollow cathode lamp of the element (nutrient) which is to be determined in AAS.

5. Feed the blank solution of 0 ppm and adjust the zero in AAS then feed 0, 1, 2 and 4ppm of standard solution of particular element.

6. Feed the sample extracts one by one into AAS and note the readings in ppm.

Calculation

1. Weight of soil = 10 g
2. Volume of extractant = 20 mL

$$\text{Dilution factor (d.f.)} = \frac{\text{Volume of extractant}}{\text{Weight of soil}} = \frac{20}{10} = 2$$

Micronutrient concentration in soil:

$$\text{ppm of micronutrient in soil} = R \times \text{d.f.} = R \times 2$$

Where,
R= ppm value obtained from sample reading (AAS reading)

Precautions

1. Use only glass-distilled or double-distilled water throughout the analysis, as ordinary water may contain trace amounts of micronutrients (Fe, Mn, Cu, Zn) which can lead to erroneous results.
2. The soil sample should be ground using a wooden pestle and mortar or a stainless steel grinder. Avoid using iron or brass equipment, as they may contaminate the sample with micronutrients.
3. The glassware used should be of high quality, preferably Corning or Borosil, to prevent leaching of impurities and ensure accuracy in measurements.
4. All glassware must be thoroughly cleaned before use. First wash with chromic acid solution ($K_2Cr_2O_7 + H_2SO_4$) to remove organic and metallic impurities, then rinse with tap water, followed by distilled water, and finally with double-distilled water. If possible, pipettes and burettes should be kept dipped in chromic acid solution before use.
5. Avoid contamination of samples and reagents during handling. Use clean, dry apparatus and do not touch the inner surfaces of glassware or containers.
6. Prepare reagents freshly or use properly stored standard solutions to maintain their strength and reliability.
7. During shaking and extraction, maintain uniform shaking time and speed to ensure consistent extraction of micronutrients.
8. Filter the extract properly using clean filter paper to obtain a clear solution before analysis in AAS.
9. Ensure proper calibration of the Atomic Absorption Spectrophotometer (AAS) using standard solutions before taking sample readings.
10. Use separate pipettes and containers for different reagents to avoid cross-contamination.
11. Perform a blank determination to correct for any impurities present in reagents or water.
12. Store prepared extracts in clean, tightly closed bottles to prevent contamination or evaporation before analysis.

Results:

The soil contains available zinc_____ppm

(Soil having available zinc less than 0.6 ppm is rated as deficient in zinc.)

The soil contains available manganese_____ ppm

(Soil having available manganese less than 2.0 ppm is rated as deficient in manganese)

The soil contains available copper_____ppm

(Soil having available copper less than 0.65 ppm is rated as deficient in copper.)

The soil contains available iron_____ppm

(Soil having available iron less than 4.5 ppm is rated as deficient in iron.)

EXPERIMENT- 8

OBJECTIVE: DETERMINATION OF TOTAL NITROGEN IN PLANT (KJELDAHL METHOD)

Principle

The total Nitrogen in soil includes the organic N compounds like proteins, amino acids and other derivatives and all forms of inorganic N, like NH_4 -N, NO -N and also NH_2-N. For the determination of total nitrogen in soil, specific method is to be adopted for getting the total nitrogen value depending on the form of N present in a particular soil sample. The organic N components can be converted into inorganic ammonia Cal salt by digestion with sulphuric acid, for reducing nitrates into ammonia Cal form, use of salicylic acid or Devarda's alloy is made in the modified Kjeldahl method. At the end of digestion, all organic and inorganic salts are converted into ammonium form which is distilled and the liberated ammonia is condensed and absorbed in boric acid, which is titrated against standard acid.

The precision of the method depends upon complete conversion of organic N into NH_4 - N, the digestion temperature and time, solid: acid ratio and the type of catalyst used have an important bearing on the method. The ideal temperature for digestion is 320-370° C. At lower temperature, the digestion may not be complete, while above 410° C, the loss of NH_3 may occur. The salt acid (weight: volume) ratio should not be less than 1:1 at the end of digestion. Commonly used catalysts to hasten the digestion process are $CuSO_4$ or Hg. Potassium sulphate is added to raise the boiling point of the acid so that loss of acid by volatilization is prevented.

Apparatus:

- Kelplus digestion and distillation unit.
- Distillation tube
- Conical flasks
- Burettes
- Pipettes etc.

Reagents:

Sulphuric acid H_2SO_4 (93-98%)

- **40% sodium hydroxide solution:** Dissolve 350 g solid NaOH in water and dilute to one litre.
- **% Boric acid:** Dissolve 40 g of boric acid powder in warm water by stirring and dilute to one litre
- **Mixed Indicator:** Dissolve 0.066 g of methyl red and 0.099 g of bromocresol green in 100 mL. of ethyl alcohol. Add 20 ml. of this mixed indicator to each litre of 4% boric acid solution.

- **0.2N H_2SO_4:** Prepare approximately 0.2N acid solution and standardize against 0.1N sodium carbonate
- **Salicyclic acid or Devarda's alloy:** Using for reducing NO, to NHa", if present in the sample.
- **Digestion mixture:** 10 g Potassium sulphate (K_2SO_4) or anhydrous sodium sulphate (AR grade) +1 g catalyst mixture (20 parts of Copper sulphate [$CuSO_4.H_2O$ (AR grade)] powder and I part metallic selenium powder. Or

10 g Potassium sulphate (K_2SO_4) or anhydrous sodium sulphate (AR grade) + 1 g of Copper sulphate [$CuSO_4.H_2O$ (AR grade)] powder.

Procedure:

- Take 1 g soil or 0.5 g plant/seed sample in distillation tube.
- Add 3-5 g digestion mixture and add 10 mL concentratedH_2SO_4.
- Heat low temperature for 30 min. and then increased heat up to 410 °C and digest for 1-2 h or until solution is crystal clear green colour and then stop the digestion. If necessary, add small amount of paraffin or glass beads to reduce frothing.
- Remove the flask from the heater and cool, add 50 ml. water and transfer to distilling flask for distillation.
- At the time of distillation, add 30 mL of 40% NaOH in the distilling flask in such a way that the contents do not mix.
- The contents are distilled for 5 minutes by pressing run bottom of kelplus distillation unit and the liberated ammonia collected in a conical flask (250 mL) containing 20 mL of 4% boric acid solution with mixed indicator.
- First remove the conical flask containing distillate and then remove the distillation tube to avoid back suction.
- Titrate the distillate against 0.2N H_2SO_4 taken in burette until pink colour starts appearing.
- Carefully remove the distillation tube and drain the contents in the sink.

If brown colour not appearing in distillation tube at the time of distillation (after 3 min) then add 5-10 ml. of 40% NaOH solution

Precautions:

1. The material after digestion should not solidify.

2. No NH4 should be lost during distillation.

3. If the indicator changes colour during distillation, determination must be repeated using either a smaller sample weight or a larger volume of standard acid.

4. Standard the 0.2N H_2SO_4 by using standard solution.

5. Check the tap water and distilled water.

6. Dip the delivery tube end in the receiver containing standard boric acid solution before adding NaOH solution in the distillation flask.

7. Close the distillation tube to the distillation apparatus tightly then add NaOH to avoid the loss of ammonia.

8. During distillation, first remove the receiver flask and then distillation tube.

9. Collect about 100 mL. of distillate in 10 minutes steady distillation.

Observation:

SI	Initial burette reading	Final burette reading	mL of (0.2N H_2SO_4)

Calculation

$$\text{N}(\%) = \frac{(A - B) \times N \times 14 \times 100}{W \times 1000}$$

Where,

- A= Volume of **0.2 N H_2SO_4** used for titration of the soil sample (mL)
- B= Volume of **0.2 N H_2SO_4** used in blank titration (mL)
- $(A|B)$= Volume of acid used to neutralize ammonia (mL)
- N= Normality of H_2SO_4 (= 0.2)
- W= Weight of soil sample (g)

Protein Content:

Crude protein was determined by multiplying percentage of nitrogen content in seeds of crops with a factor of 6.25 (Tai and Young 1974).

% CrudeProtein = % N x 6.25

EXPERIMENT - 9

OBJECTIVE: DETERMINATION OF PHOSPHORUS IN PLANTS (JACKSON, 1973)

Principle

Orthophosphates obtained after digestion with triacid react with ammonium molybdate and ammonium vanadate in a nitric acid medium to form a yellow-coloured complex. The colour develops within about 30 minutes and remains stable for 2 to 8 weeks. The intensity of the yellow colour is measured at 470 nm using a spectrophotometer.

Apparatus

- Spectrophotometer
- Volumetric Flasks
- Pipettes
- ConicalFlasks
- Digestion Flasks
- 50 ml and 1 L beakers.

Reagents

1. **Triacid mixture:**

   Prepare by mixing nitric acid (HNO_3), sulphuric acid (H_2SO_4), and perchloric acid ($HClO_4$) in the ratio of 10:1:4. For example, take 100 mL HNO_3, add 10 mL concentrated H_2SO_4, mix thoroughly, and then add 40 mL of 60% $HClO_4$.

2. **Ammonium molybdate–ammonium vanadate solution:**

   Dissolve 22.5 g of ammonium molybdate [$(NH_4)_6Mo_7O_{24}\cdot 4H_2O$] in 400 mL distilled water. In another beaker, dissolve 1.25 g ammonium vanadate in 300 mL distilled water by heating. Mix both solutions, allow to cool, then add 250 mL concentrated HNO_3 and dilute to 1 litre.

3. **Standard phosphate solution:**
   Dissolve 0.2195 g of AR-grade KH_2PO_4 in distilled water and make the volume up to 1 litre. This provides a solution containing 50 ppm phosphorus.

Procedure

1. Weigh 0.5–2.0 g of dried and finely ground plant sample into a 500 mL conical flask. Add 5 mL concentrated HNO_3 per gram of sample and leave overnight for pre-digestion.
2. Add 5 mL of triacid mixture to the pre-digested sample (add an additional 4 mL for each extra gram if the sample exceeds 5 g).
3. Digest the sample at 180–200°C until white fumes appear and most of the liquid evaporates. A clear and white residue indicates complete digestion. If the residue remains brown, add more triacid and continue digestion.
4. Transfer the digested sample to a 100 mL volumetric flask and make up the volume with distilled water.
5. Pipette 10 mL of this extract into a 50 mL volumetric flask. Add 10 mL of ammonium molybdate–ammonium vanadate reagent and dilute to volume. Allow 15–20 minutes for yellow colour development.
6. Prepare standard solutions by pipetting 0, 2, 4, 6, 8, and 10 mL from the 50 ppm stock solution into separate 50 mL volumetric flasks. Add 10 mL reagent to each and dilute to volume to obtain 0–10 ppm phosphorus standards.
7. Measure absorbance at 470 nm using a spectrophotometer. Plot a standard curve with phosphorus concentration on the X-axis and absorbance on the Y-axis. Determine the phosphorus content of the sample from this curve.
8. Take the readings of plant samples in the same manner as the standards. Calculate the ppm content of phosphorus (P) using the standard curve.

Observations and Calculation

1. Weight of plant sample = *X*g
2. Volume of digested extract prepared = 100 mL
3. Volume of digested extract taken = 10 mL
4. Volume of final coloured solution prepared = 50 mL

Calculation of Phosphorus

$$\text{P (ppm)} = R \times \frac{\text{Digested volume}}{\text{Weight of sample}} \times \frac{\text{Final volume}}{\text{Aliquot taken}} = R \times \frac{100}{X} \times \frac{50}{10} = R \times \frac{500}{X}$$

Percentage of Phosphorus

$$\text{P}\,(\%) = \frac{R \times 500}{X} \times \frac{1}{10000} = \frac{R}{20X}$$

Precautions

1. While preparing the triacid mixture, concentrated HNO_3 and $HClO_4$ should not be mixed directly. First add H_2SO_4 to HNO_3, and then add $HClO_4$ to avoid explosion.
2. The temperature during digestion should not exceed 230°C.
3. The phosphorus reading should be taken 30 minutes after the development of yellow colour.
4. The ammonium molybdate–ammonium vanadate solution should be stored in a coloured bottle to prevent oxidation.
5. After digestion, the residue should be clear and white. If digestion is incomplete, add 5 mL of triacid mixture and continue digestion.

EXPERIMENT- 10

OBJECTIVE: DETERMINATION OF POTASSIUM IN PLANT

Principle

Potassium content in plant extract is determined by flame photometry i.e. when atoms of potassium are excited in flame emit a flame of specific wave length, the intensity of emission is proportional to the concentration of K which is determined in flame photometer using K filter.

Apparatus

- Flame photometer
- Volumetric Flask
- Pipette
- Beaker
- Conical Flask 50 Ml
- Hot plate etc.

Reagents

- **Triacid mixture:**

  Prepare by mixing nitric acid (HNO_3), sulphuric acid (H_2SO_4), and perchloric acid ($HClO_4$) in the ratio of 10:1:4. For example, take 100 mL HNO_3, add 10 mL concentrated H_2SO_4, mix thoroughly, and then add 40 mL of 60% $HClO_4$.

- **Potassium standard solution:**

  Dissolve 1.9103 g of AR grade KCI in distilled water. Transfer it to 1 litre volumetric flask and make the volume. This solution contains 1000 ppm of K. To prepare 100 ppm solution take 10 ml of 1000 ppm K solution in a 100 ml volumetric flask and make up its volume.

Procedure

1. Digest the plant sample with triacid as per the method given in P determination in plant sample and make the volume in 100 ml volumetric flask.

2. Prepare 0, 2, 4, 6, 8 and 10 ppm K solution by taking 0, 2, 4, 6, 8 and 10 ml of 100 ppm K solution in 100 mi flask, respectively and making up their volume.

3. Feed the standard in flame photometer and take the reading. Plot the concentration on X-axis and flame photometer reading on Y-axis on graph paper and prepare a standard curve.

4. Feed the unknown sample in the flame photometer, note the readings and find out the concentration with the help of standard curve. If the readings of sample go above the standard then dilute the extract 5 to 10 times or more as required.

Observations and Calculation

- Weight of plant sample = *X*g
- Volume of extract prepared = 100 mL

Calculation of Potassium (K)

$$\text{K (ppm)} = R \times \frac{\text{Volume of extract prepared}}{\text{Weight of sample}} = R \times \frac{100}{X}$$

Percentage of Potassium

$$\text{K (\%)} = R \times \frac{100}{X} \times \frac{1}{1000} = \frac{R}{10X}$$

Where:

- *R*= ppm K obtained from standard curve
- *X*= weight of plant sample (g)

Precautions

1. Extract should be clear and it should be prepared from double distilled water otherwise it clogs the sucking capillary.

2 The air pressure should be maintained steadily at 0.6 kg/cm² to get uniform flame and proper automization.

3. Do not feed K solutions having higher concentration than that prescribed for the instrument.

4. Be sure that the filter used in flame photometer is of potassium.

5. After taking 8-10 readings feed distilled water and then again take the sample readings.

6. For preparing standard curve first feed the highest the concentration solution and then feed the standards of decreasing concentration of K.

7. After completion of reading close the gas then after some time stop the air supply.

EXPERIMENT -11

OBJECTIVE: DETERMINATION OF TOTAL SULPHUR IN SEEDS/PLANT

Seeds/plant were digested using the method described in 3.8.2 and sulphur estimated by Turbidimetric method by using VIS-spectrophotometer at 490 nm (Chesnin and Yien 1950) and expressed the concentration in percentage.

Apparatus:

- Spectrophotometer
- Võlumetric Flask
- Pipette
- Conical Flasks
- Beaker Etc…

Reagents:

Triacid mixture:

Prepare by mixing nitric acid (HNO_3), sulphuric acid (H_2SO_4), and perchloric acid ($HClO_4$) in the ratio of 10:1:4. For example, take 100 mL HNO_3, add 10 mL concentrated H_2SO_4, mix thoroughly, and then add 40 mL of 60% $HClO_4$.

Sodium acetate acetic acid buffer (pH 4.8) [CH_3COONa-CH_3COOH):

1000 mL. volumetric flask add 100 g sodium acetate, add distilled water 500mL to mixed, add 30 mL. 99.5% acetic acid, shake to dissolved sodium acetate the makeup volume 1000mL mark.

0.25% Gum acacia solution:

Dissolve 0.25g of chemically pure gum acacia powder in 100 mL. of hot water and filter in hot condition through Whatman No.42 filter paper. Cool and and keep in refrigerator.

Barium chloride crystal AR grade ($BaCl_2$.2H:O):

Pass AR grade BaCl: salt through 1 mm sieve and store for use.

100 ppm S solution:

Take 0.5434 g of K_2SO_4 and add 500ml. distilled water to dissolved then make up to 1000 mL. mark of volumetric flask.

Procedure:

- After digestion (see digestion of Seed and Plant samples), Transfer 10 mL of digest filtrate in to 25 mL volumetric flask and gradually add 10 ml. of sodium acetate acetic acid buffer.
- Add 1 g BaCl, $2H_2O$ powder and shake well.
- Then add 1 mL. gum acacia and make up the volume with distilled water.
- Run a blank without soil in similar manner.
- Measure the turbidity intensity at 490nm
- Run a blank side by side.

Preparation of standard curve:

Put 0, 1.25 2.5, 5.0, 7.5, 10.0, 12.5, 15.0 ml. of the working standard solution (10 mg S/litre) into

a series of 25 mL. volumetric flasks to obtain 0, 0.5, 1.0, 2.0,3.0, 4.0, 5.0 and 6.0 ppm. Develop turbidity as described above for sample aliquots. Read the turbidity intensity and prepare the curve by plotting readings against sulphur concentrations.

Precautions:

1. The temperature of the digestion mixture during digestion should not exceed 230°C.

2. The reading of the S in solution should be taken within 30 minutes from the development of turbidity.

3. A clear and white residue in flask should remain after digestion. In case of incomplete digestion the material should be again digested after addition of 5 mL of tri acid.

Observation:

S. N	Spectrophotometer reading (Abs)

Calculation:

Sulphur content in plant sample:

$$S\,(\text{mg / kg}) = R \times \text{d.f.}$$

Where:

- R= ppm reading of plant sample from standard curve
- d.f. = dilution factor

Dilution Factor (d.f.)

$$\text{d.f.} = \frac{\text{Volume of extractant}}{\text{Weight of sample}} \times \frac{\text{Final volume of solution}}{\text{Volume of aliquot taken}} = \frac{100}{0.5} \times \frac{25}{10} = 200 \times 2.5 = 500$$

Given:

- Weight of plant sample = 0.5 g
- Volume of digested extract prepared = 100 mL
- Volume of digested extract taken = 10 mL
- Volume of final coloured extract prepared = 25 mL

Note

$$\% \, \boldsymbol{S} = \frac{\boldsymbol{S \, (\text{ppm})}}{10000}$$

Nutrient Uptake

The total uptake of sulphur (or any nutrient) by the crop is calculated as:

$$\text{Nutrient uptake (kg / ha)} = \frac{\%\ \text{nutrient concentration} \times \text{Yield (kg / ha)}}{100}$$

Note

- Nutrient concentration is expressed in percentage (%)
- Yield is expressed in kg/ha
- The result gives nutrient uptake in kg/ha

EXPERIMENT- 12

OBJECTIVE: DETERMINE ORGANIC CARBON AND ORGANIC MATTER CONTENT OF A SOIL SAMPLE

Principle

A known weight of soil is treated with an excess volume of standard $K_2Cr_2O_7$ in the presence of conc. H_2SO_4. The soil is slowly digested at a low temperature by heat of dilution of H_2SO_4 and the organic carbon in the soil is thus oxidized to CO_2 The excess of K_2Cr_2O not reduced (unused) by the organic matter is titrated back against a standard solution of ferrous ammonium sulphate in the presence of H_3PO_4 acid and diphenyl amine indicator. At the end point colour of suspension changes from violet to bright green,

Reactions

1. Oxidation by Potassium Dichromate

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$$

2. Oxidation of Ferrous Sulphate

$$2FeSO_4 + H_2SO_4 + [O] \rightarrow Fe_2(SO_4)_3 + H_2O$$

3. Overall Reaction

$$K_2Cr_2O_7 + 7H_2SO_4 + 6FeSO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Fe_2(SO_4)_3 + 7H_2O$$

Reagents

1. 1 N Potassium Dichromate ($K_2Cr_2O_7$) Solution- Dissolve 49.04 g of A.R. grade $K_2Cr_2O_7$ in distilled water and dilute the volume to 1 litre.

2. N/2 Ferrous Ammonium Sulphate (FAS) Solution $\left[Fe(NH_4)_2(SO_4)_2 \cdot 6H_2O\right]$- Dissolve 392 g of FAS in distilled water. Add 15 mL concentrated H_2SO_4 and make the volume up to 2 litres with distilled water.

3. Concentrated H_2SO_4 containing 1.25% Silver Sulphate (Ag_2SO_4)- Prepare concentrated sulphuric acid containing 1.25% silver sulphate.

4. Orthophosphoric Acid (H_3PO_4, 85%) - Use 85% orthophosphoric acid as such.

5. Diphenylamine Indicator -Dissolve 0.5 g diphenylamine in a mixture of 20 mL distilled water and 100 mL concentrated H_2SO_4.

Procedure

1. Take 2 g soil in a 500 ml conical flask.

2 Add 10 ml 1 N K_2Cr_2Or solution and shake to mix.

3 Add 20 ml conc. H_2SO_4 containing 1.25% Ag:SO, and swirl the flask 2 to 30 times.

4. Allow the flask to stand for 30 minutes on an asbestos sheet for complete reaction.

5 Pour 200 ml distilled water to the flask to dilute the suspension.

6. Add 10 ml 85% H_3PO_4 and few drops of diphnylamine indicator.

7. Titrate the solution with N/2 F.A.S. till the colour changes from violet to bright green.

8. Note the volume of ferrous ammonium sulphste (F.A.S.).

9 Carry out a blank titration (without soil) in a similar manner.

Observations

1. Weight of soil = Wg
2. Volume of N/2 $Fe(NH_4)_2(SO_4)_2$used = BmL *(Blank titration)*
3. Volume of N/2 $Fe(NH_4)_2(SO_4)_2$used = SmL *(Soil sample titration)*

Calculation

$$1 \text{ mL of 1 N } K_2Cr_2O_7 = 0.003 \text{ g organic carbon}$$

Percentage of Organic Carbon

$$\% \text{ Organic Carbon} = \frac{(B - S) \times 0.003 \times 100}{2 \times W}$$

Percentage of Organic Matter

$$\% \text{ Organic Matter} = \% \text{ Organic Carbon} \times 1.724$$

Interpretation of Results (Organic Carbon)

- Below 0.50% → Low
- 0.50 – 0.75% → Medium
- Above 0.75% → High

EXPERIMENT -13

OBJECTIVE: DETERMINATION OF TOTAL NITROGEN IN UREA AND FYM SAMPLES (KJELDAHL METHOD)

Principle

The total Nitrogen in soil includes the organic N compounds like proteins, amino acids and other derivatives and all forms of inorganic N, like NH_4-N, NO_3 -N and also NH_2 -N. For the determination of total nitrogen in soil, specific method is to be adopted for getting the total nitrogen value depending on the form of N present in a particular soil sample. The precision of the method depends upon complete conversion of organic N into NH_4-N, the digestion temperature and time, solid acid ratio and the type of catalyst used have an important bearing on the method. The ideal temperature for digestion is 320°-370° C. At lower temperature, the digestion may not be complete, while above 410° C, the loss of NH3 may occur. Commonly used catalysts to hasten the digestion process are CuSO4 or Hg. Potassium sulphate is added to raise the boiling point of the acid so that loss of acid by volatilization is prevented.

Apparatus:

- Kelplus digestion and distillation unit.
- Distillation tube
- Conical flasks
- Burettes
- Pipettes etc.

Reagents:

- **Sulphuric acid**-H_2SO_4 (93-98%)
- **40% sodium hydroxide solution:** Dissolve 350 g solid NaOH in water and dilute to one litre. **4% Boric acid:** Dissolve 40 g of boric acid powder in warm water by stirring and dilute to one litre.
- **Mixed Indicator:** Dissolve 0.066 g of methyl red and 0.099 g of bromocresol green in 100 ml. of ethyl alcohol. Add 20 mL of this mixed indicator to each litre of 4% boric acid solution. **0.2N H_2SO_4:** Prepare approximately 0.2N acid solution and standardize against 0.1N sodium carbonate
- **Salicyclic acid or Devarda's alloy:** Using for reducing NO_3, to NH_4, if present in the sample.
- **Digestion mixture:** 10 g Potassium sulphate (K_2SO_4) or anhydrous sodium sulphate (AR grade) +1 g catalyst mixture (20 parts of Copper sulphate [$CuSO_4.H_2O$ (AR grade)] powder and I part metallic selenium powder.

Or

- 10 g Potassium sulphate (K_2SO_4) or anhydrous sodium sulphate (AR grade) + 1 g of Copper sulphate [$CuSO_4.H_2O$ (AR grade)] powder.

Procedure:

- Take 0.1 g urea and for FYM take 1g sample in distillation tube.
- Add 3-5 g digestion mixture and add 10 mL. concentratedH_2SO_4-
- Heat low temperature for 30 min. and then increased heat up to 410 °C and digest for 1-2 h or until solution is crystal clear green colour and then stop the digestion. If necessary, add small amount of paraffin or glass beads to reduce frothing.
- Remove the flask from the heater and cool, add 50 mL water and transfer to distilling flask for distillation.
- At the time of distillation, add 30 mL. of 40% NaOH in the distilling flask in such a way that the contents do not mix.
- The contents are distilled for 5 minutes by pressing run bottom of kelplus distillation unit and the liberated ammonia collected in a conical flask (250 mL) containing 20 mL of 4% boric acid solution with mixed indicator.
- First remove the conical flask containing distillate and then remove the distillation tube to avoid back suction.
- Titrate the distillate against 0.2N H_2SO_4 taken in burette until pink colour starts appearing.
- Carefully remove the distillation tube and drain the contents in the sink.

*If brown colour not appearing in distillation tube at the time of distillation (after 3 min) then add 5-10 mL of 40% NaOH solution

Precautions:

1. The material after digestion should not solidify.

2. No NH_4 should be lost during distillation.

3. If the indicator changes colour during distillation, determination must be repeated using either a smaller sample weight or a larger volume of standard acid.

4. Standard the 0.2N H_2SO_4 by using standard solution.

5. Check the tap water and distilled water.

6. Dip the delivery tube end in the receiver containing standard boric acid solution before adding NaOH solution in the distillation flask.

7. Close the distillation tube to the distillation apparatus tightly then add NaOH to avoid the

loss of ammonia.

8. During distillation, first remove the receiver flask and then distillation tube.

9. Collect about 100 mL of distillate in 10 minutes steady distillation.

Observation

S.N.	Initial burette reading	Final burette reading	mL of (0.2N H_2SO_4)

Calculation

$$\mathbf{N\ (\%) = \frac{(A - B) \times N \times 0.014 \times 10^2}{Weight\ of\ sample\ (g)}}$$

Where:

- A= Volume of 0.2 N H_2SO_4 used for titration of soil sample (mL)
- B= Volume of 0.2 N H_2SO_4 used in blank titration (mL)
- $(A|B)$= Volume of acid used to neutralize ammonia (mL)
- N= Normality of H_2SO_4 (= 0.2)
- Weight of sample = W (g)